CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

DIAS FAMILY DAIRY

21330 W SECOND AVE

STEVINSON CA 95374

NOTES:

PREPARED BY:

CARDOSO AG SERVICES INC.
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Dias Family Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
21330 W Second AVE	Stevinson	Merced	95374

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1905

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X055-X120-X008-XXXX

B. OPERATORS

DIAS, ROBERT

Operator name: DIAS, ROBERT

Telephone no.: (209) 667-6566 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21330 SECOND AVE	STEVINSON	CA	95374

This operator is responsible for paying permit fees.

FANELLI, ROBIN

Operator name: FANELLI, ROBIN

Telephone no.: (209) 667-6566 (Landline) (209) 678-2860 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 486	HILMAR	CA	95324

C. OWNERS

DIAS, DULCIE

Legal owner name: DIAS, DULCIE

Telephone no.: (209) 667-6566 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
21330 SECOND AVE	STEVINSON	CA	95374

DIAS, ROBERT

Legal owner name:	DIAS, ROBERT	Telephone no.:	(209) 667-6566	
			Landline	Cellular
21330 SECOND AVE		STEVINSON	CA	95374
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

FANELLI, ROBIN

Legal owner name:	FANELLI, ROBIN	Telephone no.:	(209) 667-6566	(209) 678-2860
			Landline	Cellular
P.O. Box 486		HILMAR	CA	95324
Mailing Address Number and Street		City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	302	43	80	106	0	0
Maximum number	302	43	80	106	0	0
Average number	302	43	80	106	0	0
Avg live weight (lbs)	1,000	1,150	650	500		

Predominant milk cow breed: Jersey

Average milk production: 71 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 10,037.15 *tons per reporting period*

Total nitrogen from manure: 127,247.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 89,072.97 *lbs per reporting period*

Total phosphorus from manure: 21,255.14 *lbs per reporting period*

Total potassium from manure: 58,034.20 *lbs per reporting period*

Total salt from manure: 152,084.55 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 9,840,600 *gallons*

Total nitrogen generated: 27,039.20 *lbs*

Total phosphorus generated: 7,057.25 *lbs*

Total potassium generated: 35,699.82 *lbs*

Total salt generated: 219,031.01 *lbs*

	9,840,600	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	9,840,600	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Amaral 29	29	29	2	manure	X055-X176-X007-XXXX
Amaral 39	39	39	2	manure	X055-X176-X003-XXXX
Field 1	16	16	2	both	X055-X120-X009-XXXX X055-X120-X010-XXXX
Field 2	36	36	2	both	X055-X120-X010-XXXX
Field 3	17	17	2	both	X055-X120-X007-XXXX
Totals for areas that were used for application	137	137	10		
Totals for areas that were not used for application					
Land application area totals	137	137	10		

B. CROPS AND HARVESTS

Amaral 29

Field name: Amaral 29

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	458.20 *ton*	Dry-weight		75.0	15,200.00	4,400.00	20,900.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	120.08	34.76	165.11	908.50

Amaral 29

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	605.00 *ton*	Dry-weight		70.4	13,400.00	3,200.00	15,200.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.86	165.49	39.52	187.73	827.47

Amaral 39

Field name: Amaral 39

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	635.70 *ton*	Dry-weight		73.4	13,800.00	4,600.00	20,400.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.30	119.67	39.89	176.90	910.52

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	935.60 *ton*	Dry-weight		71.4	14,100.00	2,600.00	14,500.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.99	193.48	35.68	198.97	891.94

Field 1

Field name: Field 1

09/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 09/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	240.70 *ton*	Dry-weight		66.3	17,000.00	3,700.00	31,400.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.04	172.37	37.52	318.38	973.39

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	361.60 *ton*	Dry-weight		65.0	14,800.00	2,400.00	17,700.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.60	234.14	37.97	280.01	1,059.94

Field 2

Field name: Field 2

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	626.10 *ton*	Dry-weight		70.0	23,600.00	4,900.00	38,100.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.39	246.27	51.13	397.57	1,147.85

Field 2

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	768.10 *ton*	Dry-weight		63.2	13,100.00	2,100.00	14,400.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.34	205.71	32.98	226.13	910.80

Field 3

Field name: Field 3

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	223.95 *ton*	Dry-weight		64.4	25,400.00	5,000.00	36,100.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.17	238.24	46.90	338.60	1,041.13

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	508.10 *ton*	Dry-weight		72.6	16,300.00	2,300.00	22,700.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.89	266.97	37.67	371.80	1,277.54

NUTRIENT BUDGET

A. LAND APPLICATIONS

Amaral 29 - 09/30/2022: Oats, silage-soft dough

Field name: Amaral 29

Crop: Oats, silage-soft dough — Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	131.35	21.42	23.56	0.00	452.00 *ton*
Application event totals		131.35	21.42	23.56	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	22.53	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		22.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.49	0.00	0.00	62.76	5,193,057.00 *gal*
Application event totals		1.49	0.00	0.00	62.76	

Amaral 29 - 05/15/2023: Corn, silage

Field name: Amaral 29

Crop: Corn, silage — Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	164.38	66.45	41.97	909.36	748.00 *ton*
Application event totals		164.38	66.45	41.97	909.36	

Amaral 29 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	28.16	0.00	0.00	0.00	6,750,000.00 *gal*
Application event totals		28.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	50.42	4,171,660.00 *gal*
Application event totals		1.20	0.00	0.00	50.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	53.23	4,404,282.00 *gal*
Application event totals		1.27	0.00	0.00	53.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.39	0.00	0.00	58.49	4,839,184.00 *gal*
Application event totals		1.39	0.00	0.00	58.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	54.40	4,500,793.00 *gal*
Application event totals		1.30	0.00	0.00	54.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	24.41	0.00	0.00	0.00	5,850,000.00 *gal*
Application event totals		24.41	0.00	0.00	0.00	

Amaral 29 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.54	0.00	0.00	64.55	5,341,113.00 *gal*
Application event totals		1.54	0.00	0.00	64.55	

Amaral 39 - 09/30/2022: Oats, silage-soft dough

Field name: Amaral 39

Crop: Oats, silage-soft dough

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	131.38	21.42	23.56	0.00	608.00 *ton*
Application event totals		131.38	21.42	23.56	0.00	

11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.21	0.00	0.00	0.00	7,020,000.00 *gal*
Application event totals		7.21	0.00	0.00	0.00	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.49	0.00	0.00	62.76	6,983,767.00 *gal*
Application event totals		1.49	0.00	0.00	62.76	

Amaral 39 - 05/15/2023: Corn, silage

Field name: Amaral 39

Crop: Corn, silage

Plant date: 05/15/2023

Amaral 39 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	227.31	91.89	58.04	1,257.46	1,391.00 *ton*
Application event totals		227.31	91.89	58.04	1,257.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.32	0.00	0.00	0.00	8,100,000.00 *gal*
Application event totals		8.32	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	50.42	5,610,164.00 *gal*
Application event totals		1.20	0.00	0.00	50.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.27	0.00	0.00	53.23	5,922,999.00 *gal*
Application event totals		1.27	0.00	0.00	53.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.39	0.00	0.00	58.49	6,507,868.00 *gal*
Application event totals		1.39	0.00	0.00	58.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.30	0.00	0.00	54.40	6,052,791.00 *gal*
Application event totals		1.30	0.00	0.00	54.40	

Amaral 39 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.04	0.00	0.00	0.00	7,830,000.00 *gal*
Application event totals		8.04	0.00	0.00	0.00	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.54	0.00	0.00	64.55	7,182,876.00 *gal*
Application event totals		1.54	0.00	0.00	64.55	

Field 1 - 09/30/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 09/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.02	44.10	199.03	1,243.36	924,000.00 *gal*
Irrigation Well	Ground water	41.54	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		131.57	44.10	199.03	1,243.36	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.23	18.50	102.14	693.97	415,800.00 *gal*
Canal	Surface water	1.14	0.00	0.00	47.94	2,188,612.00 *gal*
Application event totals		103.37	18.50	102.14	741.91	

Field 1 - 05/16/2023: Corn, silage

Field 1 - 05/16/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	82.85	33.49	21.15	458.33	208.00 *ton*
Application event totals		82.85	33.49	21.15	458.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	49.85	0.00	0.00	0.00	3,240,000.00 *gal*
Application event totals		49.85	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.52	0.00	0.00	64.01	2,922,119.00 *gal*
Application event totals		1.52	0.00	0.00	64.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.36	12.69	76.36	458.18	348,600.00 *gal*
Canal	Surface water	1.51	0.00	0.00	63.52	2,899,724.00 *gal*
Application event totals		68.88	12.69	76.36	521.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	62.33	2,845,616.00 *gal*
Application event totals		1.48	0.00	0.00	62.33	

Field 1 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.74	12.39	74.52	447.14	340,200.00 *gal*
Canal	Surface water	0.92	0.00	0.00	38.76	1,769,569.00 *gal*
Application event totals		66.66	12.39	74.52	485.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	48.47	0.00	0.00	0.00	3,150,000.00 *gal*
Application event totals		48.47	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.59	0.00	0.00	66.67	3,043,345.00 *gal*
Application event totals		1.59	0.00	0.00	66.67	

Field 2 - 10/01/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.84	49.89	225.17	1,406.63	2,352,000.00 *gal*
Irrigation Well	Ground water	43.08	0.00	0.00	0.00	6,300,000.00 *gal*
Application event totals		144.93	49.89	225.17	1,406.63	

Field 2 - 10/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	110.15	19.93	110.05	747.71	1,008,000.00 *gal*
Canal	Surface water	1.14	0.00	0.00	47.94	4,924,377.00 *gal*
Application event totals		111.29	19.93	110.05	795.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.65	14.72	67.95	320.70	567,000.00 *gal*
Application event totals		81.65	14.72	67.95	320.70	

Field 2 - 05/16/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	82.85	33.49	21.15	458.33	468.00 *ton*
Application event totals		82.85	33.49	21.15	458.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	49.24	0.00	0.00	70.10	7,200,000.00 *gal*
Application event totals		49.24	0.00	0.00	70.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.52	0.00	0.00	64.01	6,574,768.00 *gal*
Application event totals		1.52	0.00	0.00	64.01	

Field 2 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.89	8.83	53.16	318.95	546,000.00 *gal*
Canal	Surface water	1.51	0.00	0.00	63.52	6,524,379.00 *gal*
Application event totals		48.41	8.83	53.16	382.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	62.33	6,402,636.00 *gal*
Application event totals		1.48	0.00	0.00	62.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.89	8.83	53.16	318.95	546,000.00 *gal*
Canal	Surface water	0.92	0.00	0.00	38.76	3,981,531.00 *gal*
Application event totals		47.82	8.83	53.16	357.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	46.16	0.00	0.00	0.00	6,750,000.00 *gal*
Application event totals		46.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.59	0.00	0.00	66.67	6,847,525.00 *gal*
Application event totals		1.59	0.00	0.00	66.67	

Field 3 - 10/01/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

Field 3 - 10/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.13	49.05	221.39	1,382.99	1,092,000.00 *gal*
Irrigation Well	Ground water	43.01	0.00	0.00	0.00	2,970,000.00 *gal*
Application event totals		143.14	49.05	221.39	1,382.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.19	17.59	97.11	659.75	420,000.00 *gal*
Canal	Surface water	1.14	0.00	0.00	47.94	2,325,400.00 *gal*
Application event totals		98.33	17.59	97.11	707.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.25	15.01	69.28	326.99	273,000.00 *gal*
Application event totals		83.25	15.01	69.28	326.99	

Field 3 - 05/16/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	82.85	33.49	21.15	458.33	221.00 *ton*
Application event totals		82.85	33.49	21.15	458.33	

Field 3 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	54.74	0.00	0.00	0.00	3,780,000.00 *gal*
Application event totals		54.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.52	0.00	0.00	64.01	3,104,752.00 *gal*
Application event totals		1.52	0.00	0.00	64.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.66	17.27	103.91	623.46	504,000.00 *gal*
Canal	Surface water	1.51	0.00	0.00	63.52	3,080,957.00 *gal*
Application event totals		93.18	17.27	103.91	686.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.48	0.00	0.00	62.33	3,023,467.00 *gal*
Application event totals		1.48	0.00	0.00	62.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.66	17.27	103.91	623.46	504,000.00 *gal*
Canal	Surface water	0.92	0.00	0.00	38.76	1,880,168.00 *gal*
Application event totals		92.59	17.27	103.91	662.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	39.10	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		39.10	0.00	0.00	0.00	

Field 3 - 05/16/2023: Corn, silage

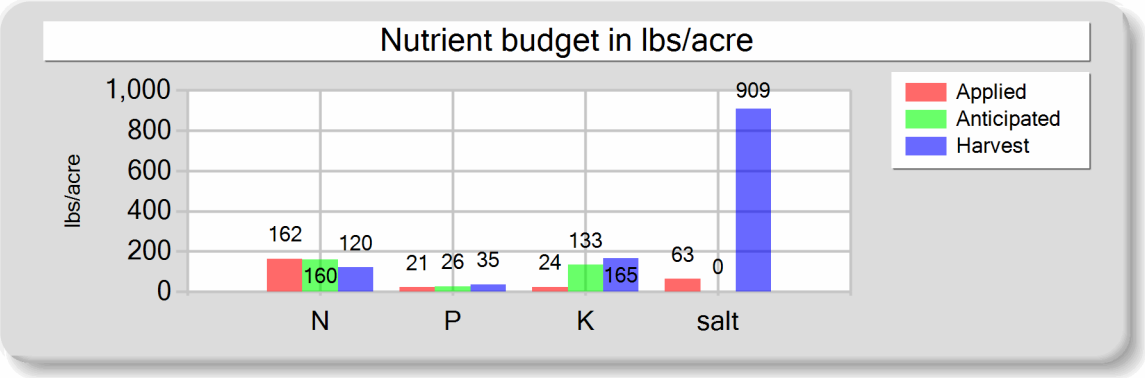
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.59	0.00	0.00	66.67	3,233,554.00 *gal*
Application event totals		1.59	0.00	0.00	66.67	

B. NUTRIENT BUDGET

Amaral 29 - 09/30/2022: Oats, silage-soft dough

Field name: Amaral 29 Crop: Oats, silage-soft dough Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.35	21.42	23.56	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	24.03	0.00	0.00	62.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.37	21.42	23.56	62.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	120.08	34.76	165.11	908.50
Nutrient balance	42.29	-13.34	-141.55	-845.74
Applied to removed ratio	1.35	0.62	0.14	0.07

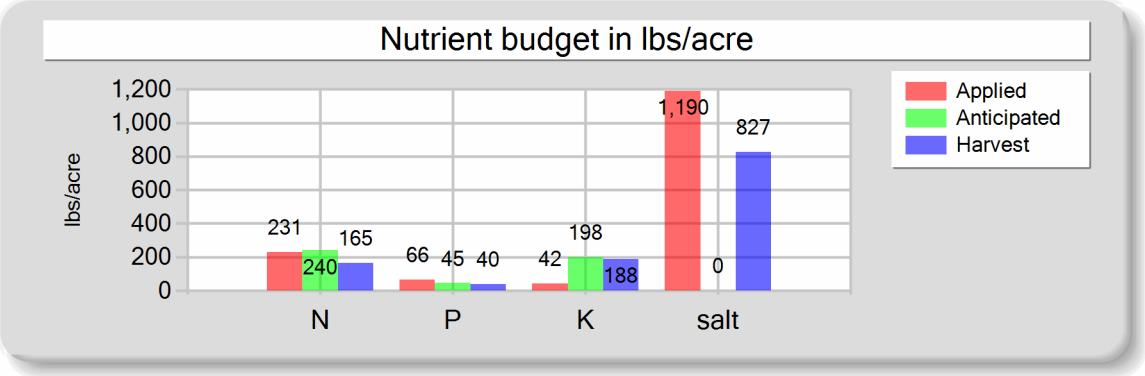
Fresh water applied	
10,593,057.00	*gallons*
390.11	*acre-inches*
13.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Amaral 29 - 05/15/2023: Corn, silage

Field name: Amaral 29 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.38	66.45	41.97	909.36
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.27	0.00	0.00	281.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.65	66.45	41.97	1,190.44
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	165.49	39.52	187.73	827.47
Nutrient balance	65.16	26.93	-145.75	362.97
Applied to removed ratio	1.39	1.68	0.22	1.44

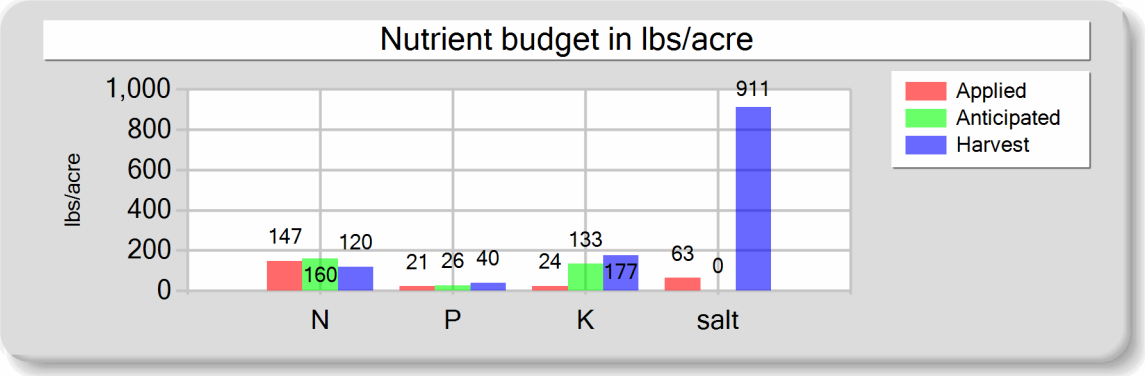
Fresh water applied	
35,857,032.00	*gallons*
1,320.49	*acre-inches*
45.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Amaral 39 - 09/30/2022: Oats, silage-soft dough

Field name: Amaral 39 Crop: Oats, silage-soft dough Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.38	21.42	23.56	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.70	0.00	0.00	62.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	147.08	21.42	23.56	62.76
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	119.67	39.89	176.90	910.52
Nutrient balance	27.41	-18.47	-153.34	-847.76
Applied to removed ratio	1.23	0.54	0.13	0.07

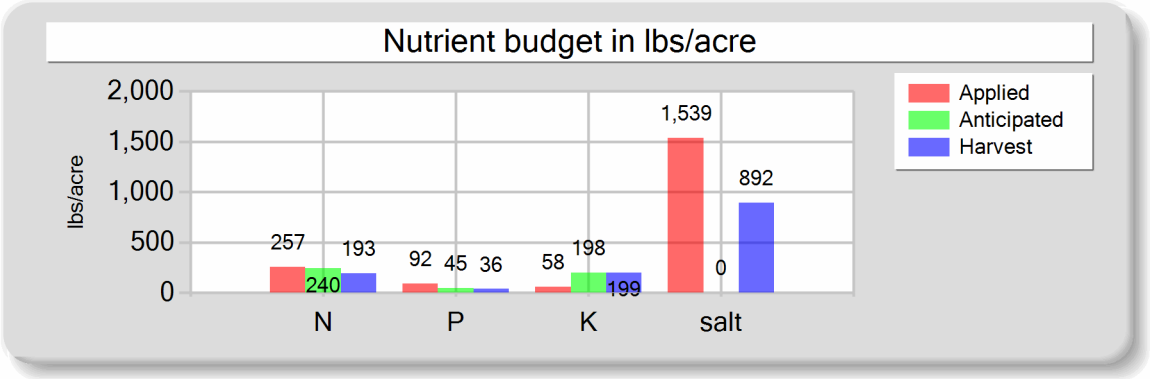
Fresh water applied	
14,003,767.00	*gallons*
515.71	*acre-inches*
13.22	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Amaral 39 - 05/15/2023: Corn, silage

Field name: Amaral 39 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	227.31	91.89	58.04	1,257.46
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	23.05	0.00	0.00	281.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.36	91.89	58.04	1,538.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	193.48	35.68	198.97	891.94
Nutrient balance	63.88	56.21	-140.93	646.61
Applied to removed ratio	1.33	2.58	0.29	1.72

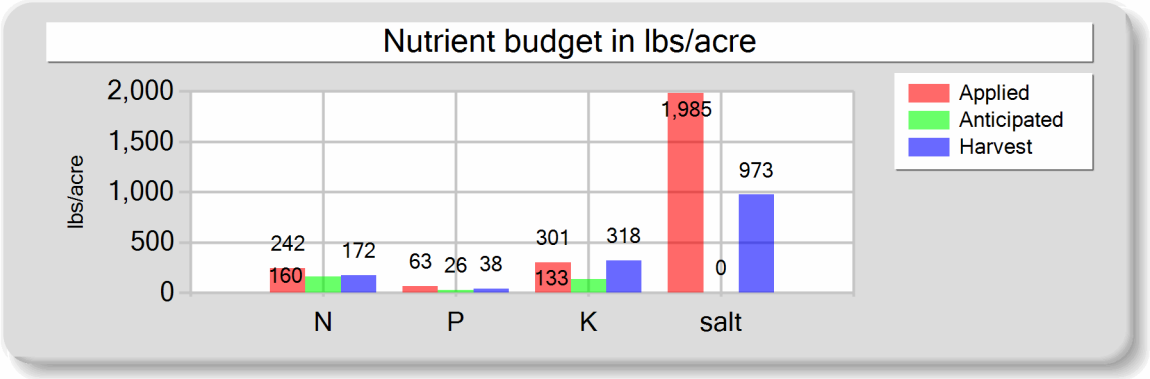
Fresh water applied	
47,206,698.00	*gallons*
1,738.46	*acre-inches*
44.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 09/30/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 09/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.25	62.59	301.18	1,937.33
Fresh water	42.68	0.00	0.00	47.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.94	62.59	301.18	1,985.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.37	37.52	318.38	973.39
Nutrient balance	69.57	25.08	-17.20	1,011.89
Applied to removed ratio	1.40	1.67	0.95	2.04

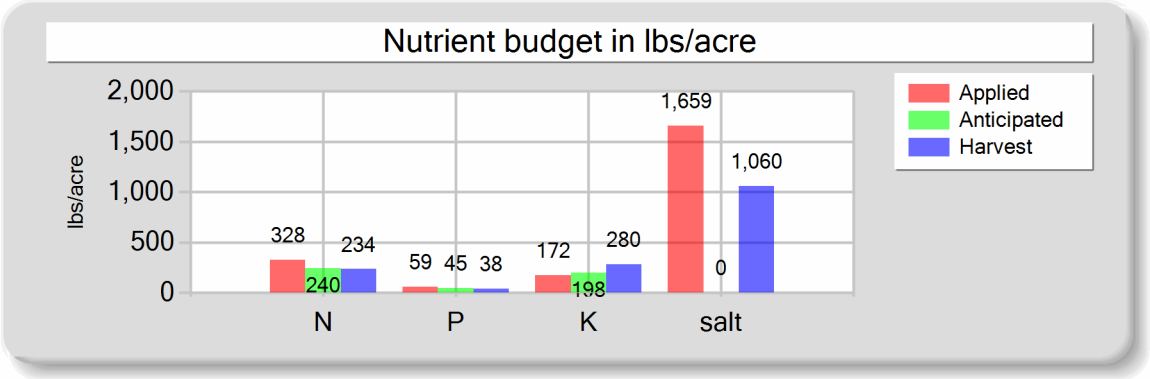
Fresh water applied	
4,888,612.00	*gallons*
180.03	*acre-inches*
11.25	*inches/acre*

Process wastewater applied	
1,339,800.00	*gallons*
49.34	*acre-inches*
3.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/16/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	82.85	33.49	21.15	458.33
Process wastewater	133.10	25.08	150.89	905.32
Fresh water	105.35	0.00	0.00	295.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.30	58.57	172.04	1,658.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	234.14	37.97	280.01	1,059.94
Nutrient balance	94.17	20.60	-107.97	599.00
Applied to removed ratio	1.40	1.54	0.61	1.57

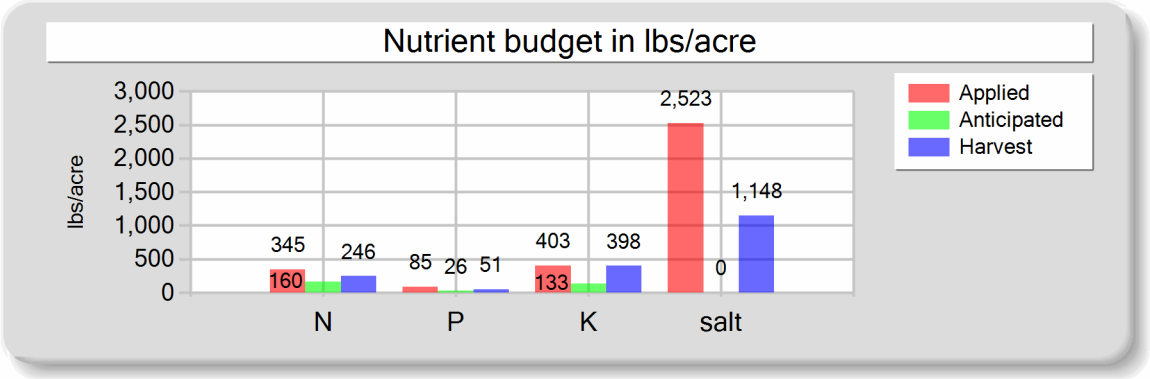
Fresh water applied	
19,870,373.00	*gallons*
731.76	*acre-inches*
45.73	*inches/acre*

Process wastewater applied	
688,800.00	*gallons*
25.37	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/01/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	293.64	84.54	403.18	2,475.04
Fresh water	44.22	0.00	0.00	47.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.86	84.54	403.18	2,522.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	246.27	51.13	397.57	1,147.85
Nutrient balance	98.60	33.41	5.60	1,375.14
Applied to removed ratio	1.40	1.65	1.01	2.20

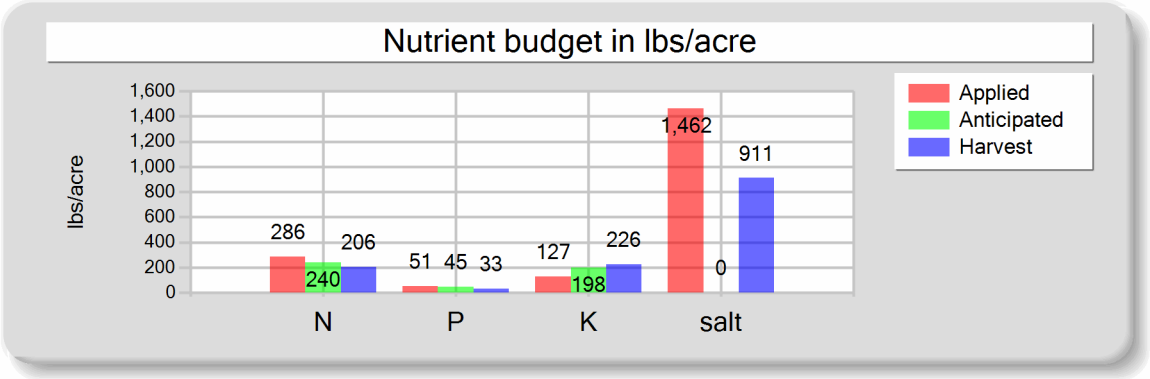
Fresh water applied	
11,224,377.00	*gallons*
413.36	*acre-inches*
11.48	*inches/acre*

Process wastewater applied	
3,927,000.00	*gallons*
144.62	*acre-inches*
4.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/16/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	82.85	33.49	21.15	458.33
Process wastewater	93.79	17.67	106.32	637.89
Fresh water	102.42	0.00	0.00	365.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.06	51.16	127.47	1,461.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	205.71	32.98	226.13	910.80
Nutrient balance	80.35	18.18	-98.66	550.82
Applied to removed ratio	1.39	1.55	0.56	1.60

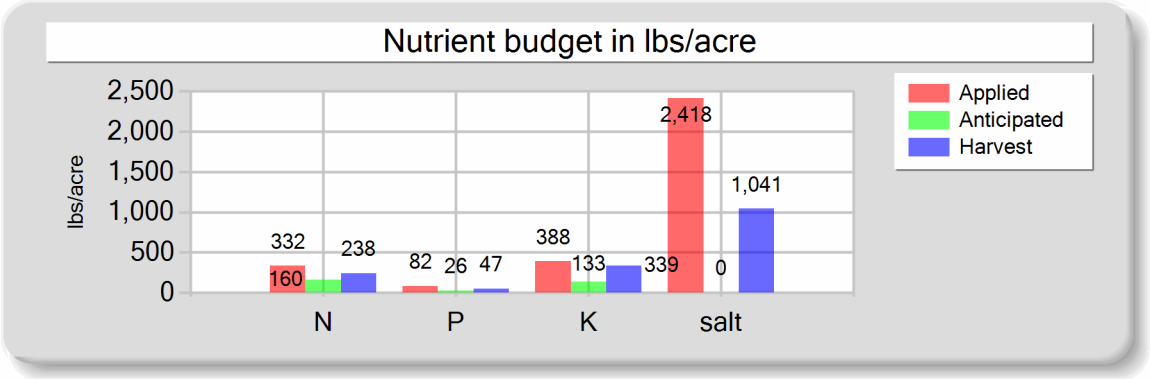
Fresh water applied	
44,280,839.00	*gallons*
1,630.71	*acre-inches*
45.30	*inches/acre*

Process wastewater applied	
1,092,000.00	*gallons*
40.21	*acre-inches*
1.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/01/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	280.57	81.64	387.78	2,369.72
Fresh water	44.15	0.00	0.00	47.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.72	81.64	387.78	2,417.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	238.24	46.90	338.60	1,041.13
Nutrient balance	93.48	34.75	49.17	1,376.54
Applied to removed ratio	1.39	1.74	1.15	2.32

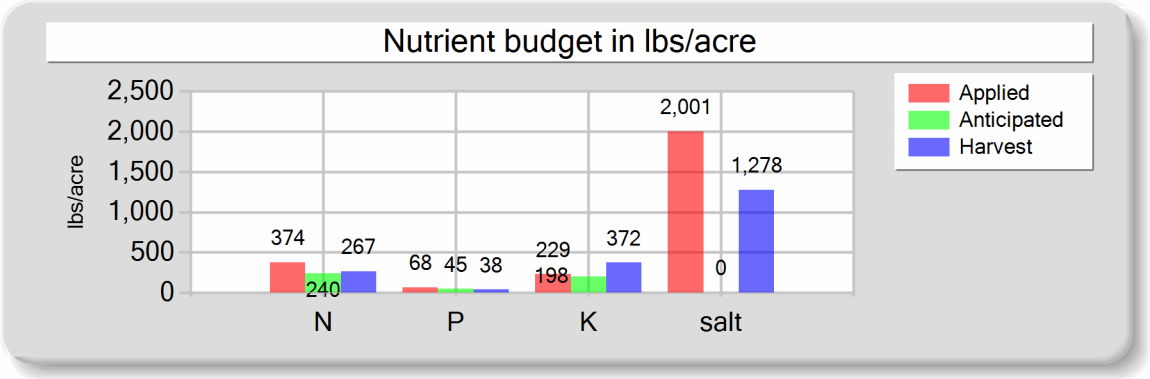
Fresh water applied	
5,295,400.00	*gallons*
195.01	*acre-inches*
11.47	*inches/acre*

Process wastewater applied	
1,785,000.00	*gallons*
65.74	*acre-inches*
3.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/16/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	82.85	33.49	21.15	458.33
Process wastewater	183.33	34.54	207.82	1,246.92
Fresh water	100.87	0.00	0.00	295.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	374.05	68.03	228.97	2,000.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	266.97	37.67	371.80	1,277.54
Nutrient balance	107.07	30.36	-142.82	723.00
Applied to removed ratio	1.40	1.81	0.62	1.57

Fresh water applied	
20,802,898.00	*gallons*
766.10	*acre-inches*
45.06	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
2.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure 2022

Sample and source description: Separator Manure 2022

Sample date: 10/17/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	3,000.00	3,300.00							
DL	0.01	0.01	0.01							

Separator Manure

Sample and source description: Separator Manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	5,700.00	3,600.00	19,100.00	2,900.00	2,300.00	3,300.00	1,000.00		7.80
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 10/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	1,700.00	3,600.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 12/06/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	185.00	166.00	0.00	1.80	91.50	413.00								5,740.00	2,580
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	223.00	0.00	1.40	85.30	471.00								5,860.00	3,200
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	620.00	181.00	0.00	1.20	112.00	517.00								6,220.00	2,440
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	370.00	152.00	0.00	0.50	69.80	420.00								5,140.00	2,520
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	331.00	244.00	0.00	0.50	76.40	590.00								6,320.00	3,450
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

Merquin Canal

Sample description: Merquin Canal

Sample date: 10/21/2020 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	0.10								50.30	42
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW#1-West

Sample description: DW#1-West

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10	39.60	6.00	200.00	290.00	0.00	54.50	181.00	1,330.00	700
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Domestic Well

DW#2-East

Sample description: DW#2-East

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10	38.40	5.20	231.00	267.00	0.00	60.70	12.30	1,510.00	830
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Irrigation Well

IW-Amaral 30

Sample description: IW-Amaral 30

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.50	0.00	14.50								1,490.00	0
DL	0.01	0.01	0.01								10.00	1

IW-Amaral 40

Sample description: IW-Amaral 40

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.80	0.00	4.80								2,610.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW-Field 2

Sample description: IW-Field 2

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	29.50	0.00	29.50								2,120.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Amaral 29 - 09/30/2022: Oats, silage-soft dough

Amaral 29 Oats

Sample and source description: Amaral 29 Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	4,400.00	20,900.00		11.50
DL	0.01	0.01	0.01		1.00

Amaral 29 - 05/15/2023: Corn, silage

Amaral 29 - 05/15/2023: Corn, silage

Corn Amaral 30

Sample and source description: Corn Amaral 30

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	3,200.00	15,200.00		6.70
DL	0.01	0.01	0.01		1.00

Amaral 39 - 09/30/2022: Oats, silage-soft dough

Amaral 39 Oats

Sample and source description: Amaral 39 Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	4,600.00	20,400.00		10.50
DL	0.01	0.01	0.01		1.00

Amaral 39 - 05/15/2023: Corn, silage

Corn Amaral 40

Sample and source description: Corn Amaral 40

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,600.00	14,500.00		6.50
DL	0.01	0.01	0.01		1.00

Field 1 - 09/30/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	3,700.00	31,400.00		9.60
DL	0.01	0.01	0.02		1.00

Field 1 - 05/16/2023: Corn, silage

Corn Field 1

Sample and source description: Corn Field 1

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	2,400.00	17,700.00		6.70
DL	0.01	0.01	0.01		1.00

Field 2 - 10/01/2022: Oats, silage-soft dough

Field 2 Oats

Sample and source description: Field 2 Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	4,900.00	38,100.00		11.00
DL	0.01	0.01	0.01		1.00

Field 2 - 05/16/2023: Corn, silage

Corn Field 2

Sample and source description: Corn Field 2

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,100.00	14,400.00		5.80
DL	0.01	0.01	0.01		1.00

Field 3 - 10/01/2022: Oats, silage-soft dough

Field 3 Oats

Sample and source description: Field 3 Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,400.00	5,000.00	36,100.00		11.10
DL	0.01	0.01	0.01		1.00

Field 3 - 05/16/2023: Corn, silage

Corn Field 3

Sample and source description: Corn Field 3

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	2,300.00	22,700.00		7.80
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

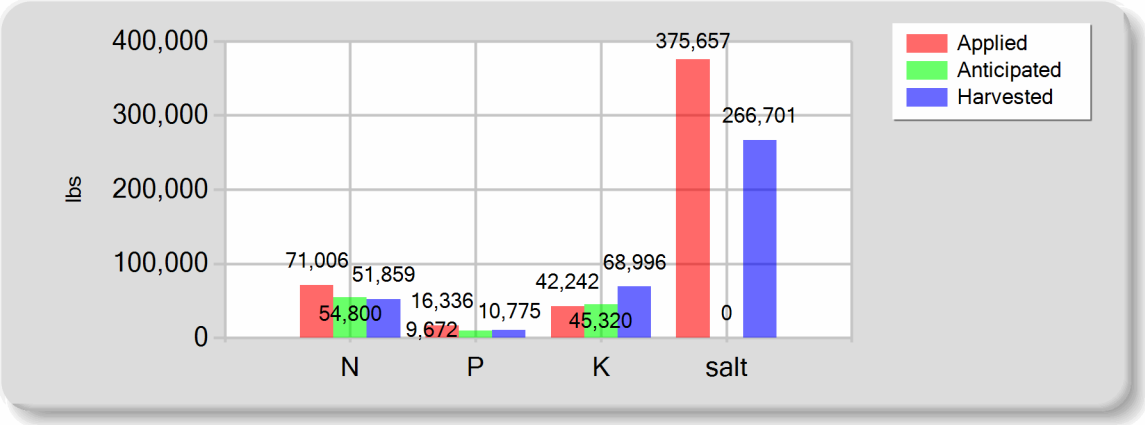
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

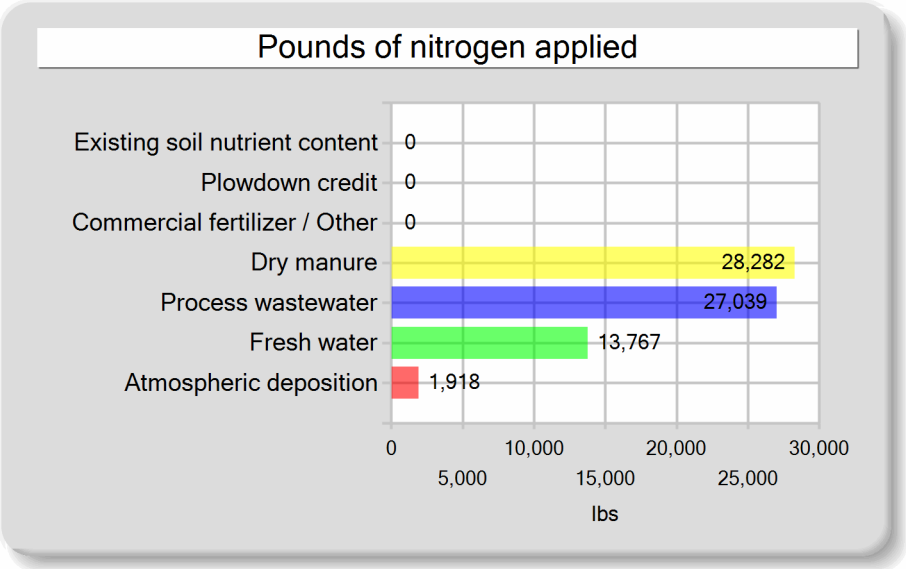
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

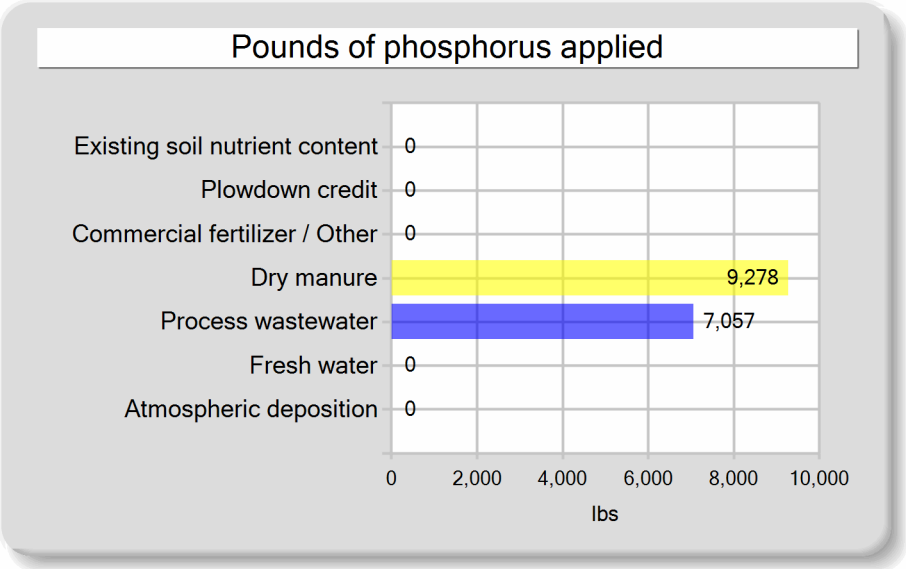
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	28,281.87	9,278.39	6,542.26	107,037.22
Process wastewater	27,039.20	7,057.25	35,699.82	219,031.01
Fresh water	13,767.15	0.00	0.00	49,588.40
Atmospheric deposition	1,918.00	0.00	0.00	0.00
Total nutrients applied	71,006.22	16,335.64	42,242.09	375,656.63
Anticipated crop nutrient removal	54,800.00	9,672.20	45,319.60	0.00
Actual crop nutrient removal	51,858.57	10,774.59	68,995.59	266,701.02
Nutrient balance	19,147.65	5,561.06	-26,753.50	108,955.61
Applied to removed ratio	1.37	1.52	0.61	1.41

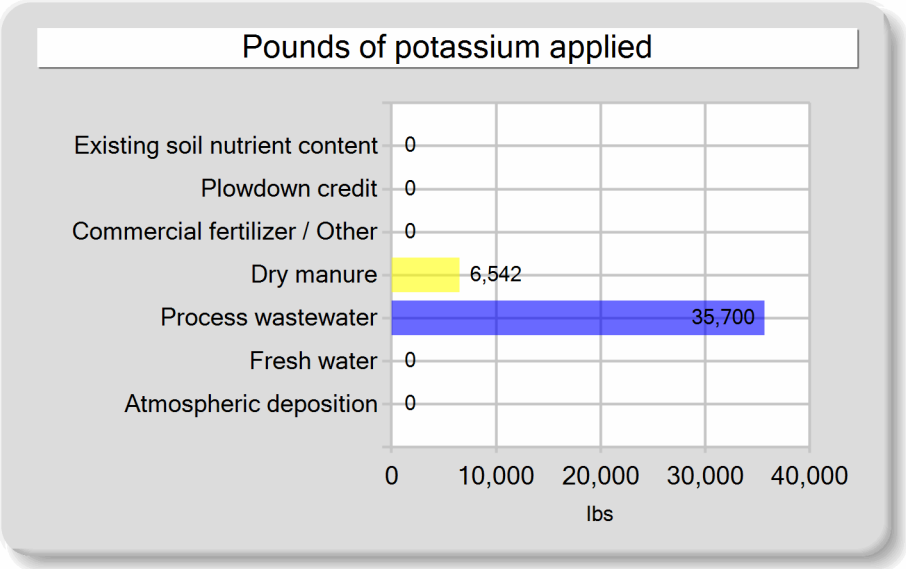
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

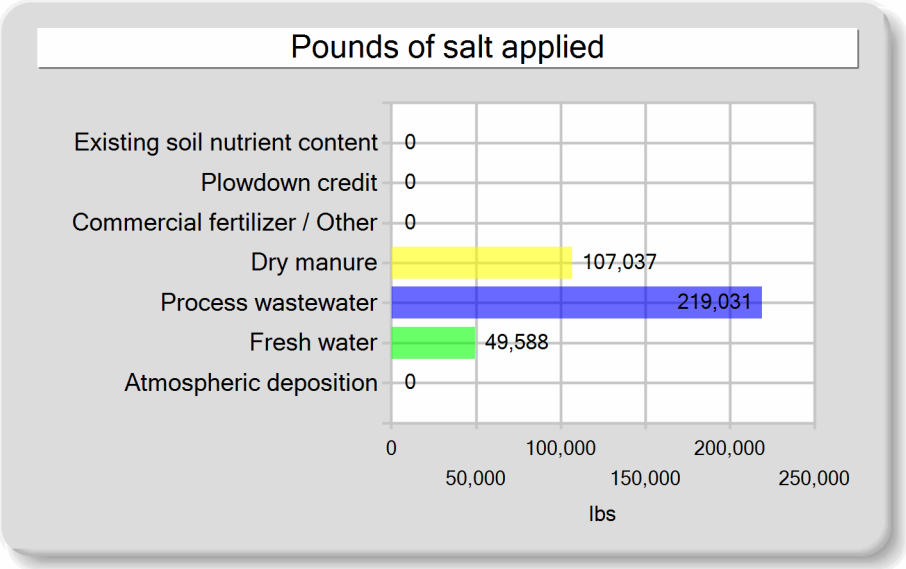


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Robin Fanelli	Robin Fanelli
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
DULCIE DIAS	ROBERT DIAS
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Dias Family Dairy
PO Box 486
Hilmar CA 95324
00-0022328 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0743
Sampled By: Robert Dias
Requested By: Robert Dias/Robin Fanelli
Submitted Date: 12/12/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW - Amaral 30	12/11/23	8:00	1490	1.49	14.5	ND		7.5
2	IW - Amaral 40	12/11/23	9:00	2610	2.61	4.8	ND		7.7
3	IW - Field 2	12/11/23	11:05	2120	2.12	29.5	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Dias Family Dairy
PO Box 486
Hilmar CA 95324
00-0022328 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0743
Sampled By: Robert Dias
Requested By: Robert Dias/Robin Fanelli
Submitted Date: 12/12/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	IW - Amaral 30	12/11/23	8:00	**1490**	**1.49**	**14.5**	39.59	3.30
2	IW - Amaral 40	12/11/23	9:00	**2610**	**2.61**	**4.8**	13.10	1.09
3	IW - Field 2	12/11/23	11:05	**2120**	**2.12**	**29.5**	80.54	6.71

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0743-01	IW - Amaral 30	Well Water	Robert Dias		12/11/2023 8:00
23L0743-02	IW - Amaral 40	Well Water	Robert Dias		12/11/2023 9:00
23L0743-03	IW - Field 2	Well Water	Robert Dias		12/11/2023 11:05

Default Cooler Temperature on Receipt °C: 0.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Sample Results

Sample: IW - Amaral 30
23L0743-01 (Water)

Sampled: 12/11/2023 8:00
Sampled By: Robert Dias

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.49**	mmhos/cm	0.01	1		12/14/23 14:59	SM 2510 B		BEL0614
Electrical Conductivity umhos	**1490**	umhos/cm	10.0	1		12/14/23 14:59	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 08:00	Field		BEL0414
Nitrate Nitrogen as NO3N	**14.5**	mg/L	0.1	1	10	12/13/23 03:31	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 14:59	SM 4500-H+	H	BEL0614
pH	**7.5**	units	1.0	1		12/14/23 14:59	SM 4500-H+	H	BEL0614

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LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Sample Results
(Continued)

Sample: IW - Amaral 40
23L0743-02 (Water)

Sampled: 12/11/2023 9:00
Sampled By: Robert Dias

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.61**	mmhos/cm	0.01	1		12/14/23 15:00	SM 2510 B		BEL0614
Electrical Conductivity umhos	**2610**	umhos/cm	10.0	1		12/14/23 15:00	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 09:00	Field		BEL0414
Nitrate Nitrogen as NO3N	**4.8**	mg/L	0.1	1	10	12/13/23 03:51	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 15:00	SM 4500-H+	H	BEL0614
pH	**7.7**	units	1.0	1		12/14/23 15:00	SM 4500-H+	H	BEL0614

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Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Sample Results
(Continued)

Sample: IW - Field 2
23L0743-03 (Water)

Sampled: 12/11/2023 11:05
Sampled By: Robert Dias

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.12**	mmhos/cm	0.01	1		12/14/23 15:01	SM 2510 B		BEL0614
Electrical Conductivity umhos	**2120**	umhos/cm	10.0	1		12/14/23 15:01	SM 2510 B		BEL0614
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:05	Field		BEL0414
Nitrate Nitrogen as NO3N	**29.5**	mg/L	0.1	1	10	12/13/23 04:10	EPA 300.0		BEL0349
Temperature	**25.0**	units	0.0	1		12/14/23 15:01	SM 4500-H+	H	BEL0614
pH	**7.8**	units	1.0	1		12/14/23 15:01	SM 4500-H+	H	BEL0614

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DELLAVALLE™ LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349										
Blank (BEL0349-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0349-BLK4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0349-BS1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.7	90-110		
LCS (BEL0349-BS2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.5	90-110		
LCS (BEL0349-BS3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.1	90-110		
Duplicate (BEL0349-DUP1)		**Source: 23L0574-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L		4.6			1.92	10
Duplicate (BEL0349-DUP2)		**Source: 23L0733-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.6		0.1	mg/L		5.6			0.659	10
Duplicate (BEL0349-DUP3)		**Source: 23L0736-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L		6.8			0.930	10
Matrix Spike (BEL0349-MS1)		**Source: 23L0574-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.7		0.1	mg/L	5.000	4.6	102	90-110		
Matrix Spike (BEL0349-MS2)		**Source: 23L0733-01**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	10.9		0.1	mg/L	5.000	5.6	106	90-110		
Matrix Spike (BEL0349-MS3)		**Source: 23L0736-02**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	11.9		0.1	mg/L	5.000	6.8	102	90-110		
Reference (BEL0349-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.8	90-110		
Reference (BEL0349-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		

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LABORATORY INC

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PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349 (Continued)										
Reference (BEL0349-SRM3)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.5	90-110		
Reference (BEL0349-SRM4)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.8	90-110		

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Received: 12/12/2023 15:51
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614										
Blank (BEL0614-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.10		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	97.4		10.0	umhos/cm						
pH	3.2		1.0	units						
Blank (BEL0614-BLK2)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	0.01		0.01	mmhos/cm						
pH	4.4		1.0	units						
Electrical Conductivity umhos	10.5		10.0	umhos/cm						
Blank (BEL0614-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Duplicate (BEL0614-DUP1)		**Source: 23L0743-03**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	2.12		0.01	mmhos/cm		2.12			0.165	10
Electrical Conductivity umhos	2120		10.0	umhos/cm		2120			0.165	10
pH	7.8		1.0	units		7.8			0.258	10
Duplicate (BEL0614-DUP2)		**Source: 23L0771-01**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.66		0.01	mmhos/cm		0.66			0.923	10
Electrical Conductivity umhos	664		10.0	umhos/cm		658			0.923	10
pH	7.5		1.0	units		7.6			0.529	10
Reference (BEL0614-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	458			umhos/cm	426.0		107	90-110		
Reference (BEL0614-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0614-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1150			umhos/cm	1000		115	90-110		
Electrical Conductivity umhos	1150			umhos/cm	1000		115	90-110		
Reference (BEL0614-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		

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LABORATORY INC

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PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/20/2023 16:33

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0614 (Continued)										
Reference (BEL0614-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEL0614-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0614-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0614-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/12/23 15:51 23L0743

JG

WATER WORK REQUEST

Bill To: Acct No. 22328 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Dias Family Dairy
Address: PO Box 486
City, State, Zip: Hilmar, CA 95324
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Robert Dias / Robin Fanelli

Ranch _____

Date sampled: 12/11/23

Sampled by: Robert Dias

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
(✓) Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. IW-Amaral 30	Sampled From:	12/11/23	8AM	Ø	0.9
2. IW-Amaral 40	Sampled From:	12/11/23	9AM	Ø	2.3
3. IW-Field 2	Sampled From:	12/11/23	11:05A	Ø	2.1
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	RDias	DiasFamilyDairy	12/11/23 11:05A	12/11/23 1125 A
Second	MPedroso	CAS INC	12/11/23 @ 11:35am	12/12/23 @9:30am
Third	JLF	DLV	12-12-23 9:30	12-12-23 2:01
Fourth	JS	DLL	12/12/23 15:51	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a late damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through calunder its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1							
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/12/23 15:51 23L0743

SEND OUT

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0688
Sampled By: M.Pedroso
Requested By: Robert Dias/Robin Fanelli
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project:
Crop ID:

Dias Family Dairy
PO Box 486
Hilmar CA 95324
00-0022328 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TDS mg/L	CO_3 as $CaCO_3$ mg/L	HCO_3 as $CaCO_3$ mg/L	Cl mg/L	SO_4 mg/L	Ca mg/L	Mg mg/L	Na mg/L	pH at 25°C unit
1	DW#1-West	12/11/23	11:20	1330	1.33	0.1	ND		700	ND	290	181	54.5	39.6	6.0	200	7.8
2	DW#2-East	12/11/23	11:25	1510	1.51	0.1	ND		830	ND	267	12.3	60.7	38.4	5.2	231	7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23L0688
Sampled By: M.Pedroso
Requested By: Robert Dias/Robin Fanelli
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project:
Crop ID:

Dias Family Dairy
PO Box 486
Hilmar CA 95324
00-0022328 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW#1-West	12/11/23	11:20	**1330**	**1.33**	0.1	0.27	0.02
2	DW#2-East	12/11/23	11:25	**1510**	**1.51**	0.1	0.27	0.02

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0688-01	DW#1-West	Ag Water	M.Pedroso		12/11/2023 11:20
23L0688-02	DW#2-East	Ag Water	M.Pedroso		12/11/2023 11:25

Default Cooler Temperature on Receipt °C: 0.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Sample Results

Sample: DW#1-West
23L0688-01 (Water)

Sampled: 12/11/2023 11:20
Sampled By: M.Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**290**	mg/L	10.0	1		12/14/23 11:24	SM 2320 B		BEL0492
Calcium	**39.6**	mg/L	0.1	1		12/18/23 10:33	EPA 200.7		BEL0570
Chloride	**181**	mg/L	0.2	1	250	12/12/23 22:34	EPA 300.0		BEL0349
Carbonate as CaCO3	ND	mg/L	1	1		12/14/23 11:24	SM 2320 B		BEL0492
Electrical Conductivity	**1.33**	mmhos/cm	0.01	1		12/14/23 11:24	SM 2510 B		BEL0492
Electrical Conductivity umhos	**1330**	umhos/cm	10.0	1		12/14/23 11:24	SM 2510 B		BEL0492
Bicarbonate as CaCO3	**290**	mg/L	5.00	1		12/14/23 11:24	SM 2320 B		BEL0492
Potassium	**3.33**	mg/L	0.500	1		12/18/23 10:33	EPA 200.7		BEL0570
Magnesium	**6.0**	mg/L	0.1	1		12/18/23 10:33	EPA 200.7		BEL0570
Sodium	**200**	mg/L	1	1		12/18/23 10:33	EPA 200.7		BEL0570
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:20	Field		BEL0381
Nitrate Nitrogen as NO3N	**0.1**	mg/L	0.1	1	10	12/12/23 22:34	EPA 300.0		BEL0349
Hydroxide as CaCO3	ND	mg/L	1.00	1		12/14/23 11:24	SM 2320 B		BEL0492
pH	**7.8**	units	1.0	1		12/14/23 11:24	SM 4500-H+	H	BEL0492
Temperature	**25.0**	units	0.0	1		12/14/23 11:24	SM 4500-H+	H	BEL0492
Sulfate (SO4)	**54.5**	mg/L	0.5	1	250	12/12/23 22:34	EPA 300.0		BEL0349
Total Filterable Solids (TDS)	**700**	mg/L	10.0	1		12/14/23 16:23	SM 2540 C		BEL0431

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LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Sample Results
(Continued)

Sample: DW#2-East
23L0688-02 (Water)

Sampled: 12/11/2023 11:25
Sampled By: M.Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Alkalinity as CaCO3	**267**	mg/L	10.0	1		12/14/23 11:31	SM 2320 B		BEL0492
Calcium	**38.4**	mg/L	0.1	1		12/18/23 10:34	EPA 200.7		BEL0570
Chloride	**12.3**	mg/L	0.2	1	250	12/18/23 17:15	EPA 300.0		BEL0510
Carbonate as CaCO3	ND	mg/L	1	1		12/14/23 11:31	SM 2320 B		BEL0492
Electrical Conductivity	**1.51**	mmhos/cm	0.01	1		12/14/23 11:31	SM 2510 B		BEL0492
Electrical Conductivity umhos	**1510**	umhos/cm	10.0	1		12/14/23 11:31	SM 2510 B		BEL0492
Bicarbonate as CaCO3	**267**	mg/L	5.00	1		12/14/23 11:31	SM 2320 B		BEL0492
Potassium	**3.58**	mg/L	0.500	1		12/18/23 10:34	EPA 200.7		BEL0570
Magnesium	**5.2**	mg/L	0.1	1		12/18/23 10:34	EPA 200.7		BEL0570
Sodium	**231**	mg/L	1	1		12/18/23 10:34	EPA 200.7		BEL0570
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:25	Field		BEL0381
Nitrate Nitrogen as NO3N	**0.1**	mg/L	0.1	1	10	12/13/23 01:13	EPA 300.0		BEL0349
Hydroxide as CaCO3	ND	mg/L	1.00	1		12/14/23 11:31	SM 2320 B		BEL0492
pH	**7.9**	units	1.0	1		12/14/23 11:31	SM 4500-H+	H	BEL0492
Temperature	**25.0**	units	0.0	1		12/14/23 11:31	SM 4500-H+	H	BEL0492
Sulfate (SO4)	**60.7**	mg/L	0.5	1	250	12/13/23 01:13	EPA 300.0		BEL0349
Total Filterable Solids (TDS)	**830**	mg/L	10.0	1		12/14/23 16:23	SM 2540 C		BEL0431

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LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349										
Blank (BEL0349-BLK1)					Prepared & Analyzed: 12/12/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0349-BLK2)					Prepared & Analyzed: 12/12/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0349-BLK3)					Prepared & Analyzed: 12/13/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
Blank (BEL0349-BLK4)					Prepared & Analyzed: 12/13/2023					
Chloride	ND		0.2	mg/L						
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Sulfate (SO4)	ND		0.5	mg/L						
LCS (BEL0349-BS1)					Prepared & Analyzed: 12/12/2023					
Chloride	4.7		0.2	mg/L	5.000		94.6	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.7	90-110		
Sulfate (SO4)	4.5		0.5	mg/L	5.000		90.1	90-110		
LCS (BEL0349-BS2)					Prepared & Analyzed: 12/13/2023					
Chloride	4.7		0.2	mg/L	5.000		94.4	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.5	90-110		
Sulfate (SO4)	4.5		0.5	mg/L	5.000		90.1	90-110		
LCS (BEL0349-BS3)					Prepared & Analyzed: 12/13/2023					
Chloride	4.8		0.2	mg/L	5.000		95.0	90-110		
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.1	90-110		
Sulfate (SO4)	4.5		0.5	mg/L	5.000		90.9	90-110		
Duplicate (BEL0349-DUP1)		**Source: 23L0574-01**			Prepared & Analyzed: 12/13/2023					
Chloride	44.7		0.2	mg/L		44.0			1.64	10
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L		4.6			1.92	10
Sulfate (SO4)	82.2		0.5	mg/L		81.0			1.51	10
Duplicate (BEL0349-DUP2)		**Source: 23L0733-01**			Prepared & Analyzed: 12/13/2023					
Chloride	54.1		0.2	mg/L		53.8			0.512	10
Nitrate Nitrogen as NO3N	5.6		0.1	mg/L		5.6			0.659	10
Sulfate (SO4)	6.3		0.5	mg/L		6.3			1.24	10

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Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0349 (Continued)										
Duplicate (BEL0349-DUP3)			**Source: 23L0736-02**		Prepared & Analyzed: 12/13/2023					
Chloride	94.2		0.2	mg/L		93.5			0.696	10
Nitrate Nitrogen as NO3N	6.9		0.1	mg/L		6.8			0.930	10
Sulfate (SO4)	90.5		0.5	mg/L		89.5			1.12	10
Matrix Spike (BEL0349-MS1)			**Source: 23L0574-01**		Prepared & Analyzed: 12/12/2023					
Chloride	49.5		0.2	mg/L	5.000	44.0	111	90-110		
Nitrate Nitrogen as NO3N	9.7		0.1	mg/L	5.000	4.6	102	90-110		
Sulfate (SO4)	86.9		0.5	mg/L	5.000	81.0	117	90-110		
Matrix Spike (BEL0349-MS2)			**Source: 23L0733-01**		Prepared & Analyzed: 12/13/2023					
Chloride	58.9		0.2	mg/L	5.000	53.8	101	90-110		
Nitrate Nitrogen as NO3N	10.9		0.1	mg/L	5.000	5.6	106	90-110		
Sulfate (SO4)	11.3		0.5	mg/L	5.000	6.3	101	90-110		
Matrix Spike (BEL0349-MS3)			**Source: 23L0736-02**		Prepared & Analyzed: 12/13/2023					
Chloride	97.8		0.2	mg/L	5.000	93.5	84.3	90-110		
Nitrate Nitrogen as NO3N	11.9		0.1	mg/L	5.000	6.8	102	90-110		
Sulfate (SO4)	94.2		0.5	mg/L	5.000	89.5	94.6	90-110		
Reference (BEL0349-SRM1)					Prepared & Analyzed: 12/12/2023					
Chloride	12.0			mg/L	12.50		95.9	90-110		
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.8	90-110		
Sulfate (SO4)	9.4			mg/L	10.00		93.9	90-110		
Reference (BEL0349-SRM2)					Prepared & Analyzed: 12/13/2023					
Chloride	12.0			mg/L	12.50		96.3	90-110		
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		
Sulfate (SO4)	9.4			mg/L	10.00		93.9	90-110		
Reference (BEL0349-SRM3)					Prepared & Analyzed: 12/13/2023					
Chloride	12.1			mg/L	12.50		96.6	90-110		
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.5	90-110		
Sulfate (SO4)	9.4			mg/L	10.00		94.3	90-110		
Reference (BEL0349-SRM4)					Prepared & Analyzed: 12/13/2023					
Chloride	12.1			mg/L	12.50		97.0	90-110		
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.8	90-110		
Sulfate (SO4)	9.4			mg/L	10.00		94.1	90-110		

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Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0431										
Blank (BEL0431-BLK1)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEL0431-BS1)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Total Filterable Solids (TDS)	23.8		10.0	mg/L	2000		1.19	0-200		
Duplicate (BEL0431-DUP1)			**Source: 23L0688-01**		Prepared: 12/13/2023 Analyzed: 12/14/2023					
Total Filterable Solids (TDS)	730		10.0	mg/L		700			4.20	10
Duplicate (BEL0431-DUP2)			**Source: 23L0744-01**		Prepared: 12/13/2023 Analyzed: 12/14/2023					
Total Filterable Solids (TDS)	1020		10.0	mg/L		1010			0.985	10
Reference (BEL0431-SRM1)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Total Filterable Solids (TDS)	397			mg/L	390.0		102	90-110		

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DELLAVALLE™ LABORATORY INC

Dias Family Dairy
PO Box 486
Hilmar, CA 95324

Account# 00-0022328
Account Manager: Ben Nydam
Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0492										
Blank (BEL0492-BLK1)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Carbonate as CaCO3	ND		1	mg/L						
Electrical Conductivity	ND		0.01	mmhos/cm						
Hydroxide as CaCO3	ND		1.00	mg/L						
Alkalinity as CaCO3	ND		10.0	mg/L						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Temperature	25.0		0.0	units						
Blank (BEL0492-BLK2)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Alkalinity as CaCO3	ND		10.0	mg/L						
Carbonate as CaCO3	ND		1	mg/L						
Hydroxide as CaCO3	ND		1.00	mg/L						
pH	5.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	units						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Blank (BEL0492-BLK3)					Prepared: 12/13/2023 Analyzed: 12/14/2023					
pH	5.5		1.0	units						
Alkalinity as CaCO3	ND		10.0	mg/L						
Electrical Conductivity	ND		0.01	mmhos/cm						
Hydroxide as CaCO3	ND		1.00	mg/L						
Carbonate as CaCO3	ND		1	mg/L						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Bicarbonate as CaCO3	ND		5.00	mg/L						
Temperature	25.0		0.0	units						
Duplicate (BEL0492-DUP1)		**Source: 23L0701-03**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
Alkalinity as CaCO3	575		10.0	mg/L		533			7.60	10
pH	7.8		1.0	units		7.9			1.53	10
Carbonate as CaCO3	ND		1	mg/L		ND				10
Electrical Conductivity	3.09		0.01	mmhos/cm		3.07			0.516	10
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
Electrical Conductivity umhos	3090		10.0	umhos/cm		3070			0.516	10
Duplicate (BEL0492-DUP2)		**Source: 23L0857-01**			Prepared: 12/13/2023 Analyzed: 12/14/2023					
pH	7.8		1.0	units		7.7			0.258	10
Hydroxide as CaCO3	ND		1.00	mg/L		ND				10
Alkalinity as CaCO3	345		10.0	mg/L		327			5.28	10
Electrical Conductivity	1.46		0.01	mmhos/cm		1.46			0.0890	10
Carbonate as CaCO3	ND		1	mg/L		ND				10

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0492 (Continued)										
Duplicate (BEL0492-DUP2)			**Source: 23L0857-01**		Prepared: 12/13/2023	Analyzed: 12/14/2023				
Electrical Conductivity umhos	1460		10.0	umhos/cm		1460			0.0890	10
Reference (BEL0492-SRM1)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
Alkalinity as CaCO3	129			mg/L	128.0		100	90-110		
Electrical Conductivity	446			umhos/cm	426.0		105	90-110		
Reference (BEL0492-SRM2)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
Alkalinity as CaCO3	128			mg/L	128.0		100	90-110		
Electrical Conductivity	455			umhos/cm	426.0		107	90-110		
Reference (BEL0492-SRM3)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
Alkalinity as CaCO3	123			mg/L	128.0		95.9	90-110		
Electrical Conductivity	447			umhos/cm	426.0		105	90-110		
Reference (BEL0492-SRM4)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0492-SRM5)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
pH	4.1			units	4.000		102	97.5-102.5		
Reference (BEL0492-SRM6)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
pH	4.0			units	4.000		99.8	97.5-102.5		
Reference (BEL0492-SRM7)					Prepared: 12/13/2023	Analyzed: 12/14/2023				
pH	7.5			units	7.520		100	67021-101.32		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0510										
Blank (BEL0510-BLK1)					Prepared & Analyzed: 12/18/2023					
Chloride	ND		0.2	mg/L						
Blank (BEL0510-BLK2)					Prepared & Analyzed: 12/18/2023					
Chloride	ND		0.2	mg/L						
Blank (BEL0510-BLK3)					Prepared & Analyzed: 12/19/2023					
Chloride	ND		0.2	mg/L						
LCS (BEL0510-BS1)					Prepared & Analyzed: 12/18/2023					
Chloride	4.7		0.2	mg/L	5.000		94.4	90-110		
LCS (BEL0510-BS2)					Prepared & Analyzed: 12/19/2023					
Chloride	4.7		0.2	mg/L	5.000		94.4	90-110		
Duplicate (BEL0510-DUP1)		**Source: 23L0774-11**			Prepared & Analyzed: 12/18/2023					
Chloride	24.8		0.2	mg/L		24.7			0.469	10
Duplicate (BEL0510-DUP2)		**Source: 23L0915-01**			Prepared & Analyzed: 12/19/2023					
Chloride	4.1		0.2	mg/L		4.1			0.610	10
Matrix Spike (BEL0510-MS1)		**Source: 23L0774-11**			Prepared & Analyzed: 12/18/2023					
Chloride	29.1		0.2	mg/L	5.000	24.7	89.4	90-110		
Matrix Spike (BEL0510-MS2)		**Source: 23L0915-01**			Prepared & Analyzed: 12/19/2023					
Chloride	9.2		0.2	mg/L	5.000	4.1	103	90-110		
Reference (BEL0510-SRM1)					Prepared & Analyzed: 12/18/2023					
Chloride	12.7			mg/L	12.50		101	90-110		
Reference (BEL0510-SRM2)					Prepared & Analyzed: 12/18/2023					
Chloride	12.7			mg/L	12.50		102	90-110		
Reference (BEL0510-SRM3)					Prepared & Analyzed: 12/19/2023					
Chloride	12.8			mg/L	12.50		102	90-110		

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Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0570										
Blank (BEL0570-BLK1)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Potassium	ND		0.500	mg/L						
Calcium	0.1		0.1	mg/L						
Sodium	ND		1	mg/L						
Magnesium	ND		0.1	mg/L						
Blank (BEL0570-BLK2)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Potassium	ND		0.500	mg/L						
Sodium	ND		1	mg/L						
Calcium	ND		0.1	mg/L						
Magnesium	ND		0.1	mg/L						
LCS (BEL0570-BS1)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Sodium	38		1	mg/L	35.71		107	90-110		
Potassium	37.0		0.500	mg/L	35.71		104	90-110		
Calcium	38.7		0.1	mg/L	35.71		108	90-110		
Magnesium	36.6		0.1	mg/L	35.71		103	90-110		
LCS (BEL0570-BS2)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Potassium	37.7		0.500	mg/L	35.71		105	90-110		
Calcium	39.6		0.1	mg/L	35.71		111	90-110		
Sodium	38		1	mg/L	35.71		105	90-110		
Magnesium	37.4		0.1	mg/L	35.71		105	90-110		
Duplicate (BEL0570-DUP1)		**Source: 23L0701-03**			Prepared: 12/13/2023	Analyzed: 12/18/2023				
Sodium	446		1	mg/L		450			1.00	15
Potassium	3.86		0.500	mg/L		3.84			0.467	15
Calcium	60.9		0.1	mg/L		61.0			0.197	15
Magnesium	20.3		0.1	mg/L		20.4			0.492	15
Matrix Spike (BEL0570-MS1)		**Source: 23L0701-03**			Prepared: 12/13/2023	Analyzed: 12/18/2023				
Sodium	450		1	mg/L	35.71	450	NR	90-110		
Calcium	96.3		0.1	mg/L	35.71	61.0	98.6	90-110		
Potassium	41.7		0.500	mg/L	35.71	3.84	106	90-110		
Magnesium	54.3		0.1	mg/L	35.71	20.4	95.1	90-110		
Matrix Spike (BEL0570-MS2)		**Source: 23L0877-04**			Prepared: 12/13/2023	Analyzed: 12/18/2023				
Sodium	60		1	mg/L	35.71	25	97.7	90-110		
Potassium	41.4		0.500	mg/L	35.71	4.33	104	90-110		
Calcium	62.2		0.1	mg/L	35.71	23.4	109	90-110		
Magnesium	44.6		0.1	mg/L	35.71	8.5	101	90-110		
Reference (BEL0570-SRM2)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Sodium	96			mg/L	91.50		105	90-110		

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LABORATORY INC

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Submitted By: Robert Dias/Robin Fanelli

Received: 12/12/2023 15:51
Reported: 12/21/2023 12:20

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0570 (Continued)										
Reference (BEL0570-SRM2)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Potassium	23.7			mg/L	21.90		108	90-110		
Reference (BEL0570-SRM3)					Prepared: 12/13/2023	Analyzed: 12/18/2023				
Calcium	49.3			mg/L	45.90		107	90-110		
Magnesium	37.1			mg/L	35.60		104	90-110		

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12/12/23 15:51 23L0688

JS

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 22328 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Dias Family Dairy
Address: PO Box 486
City, State, Zip: Hilmar, CA 95324
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Robert Dias / Robin Fanelli

Ranch _____

Date sampled: 12/11/23

Sampled by: MPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

(X) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES

		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW#1-West	Sampled From:	12/11/23	1120A	Ø	0.9
2. DW#2-East	Sampled From:	12/11/23	1125A	Ø	2.4
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPedr	CAS Inc	12/11/23 1125A	12/12/23 @ 9:30am
Second	JLF	DLV	12-12-23 9:30	12-12-23 2:01
Third				
Fourth	JS	DLV	12/12/23 15:51	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The pries will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☐ Blue Ice ☒ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1								
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/12/23 15:51 23L0688

DELIVERED SENDOUT